

CSE274 Final Project Report

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1 Introduction

Inverse rendering aims to recover the physical properties of a scene, such as geometry, materials, and lighting, from images using neural networks. The motivation behind this is to enable realistic scene understanding and editing, where we can relight scenes, change materials, or render them from new viewpoints in a physically consistent way.

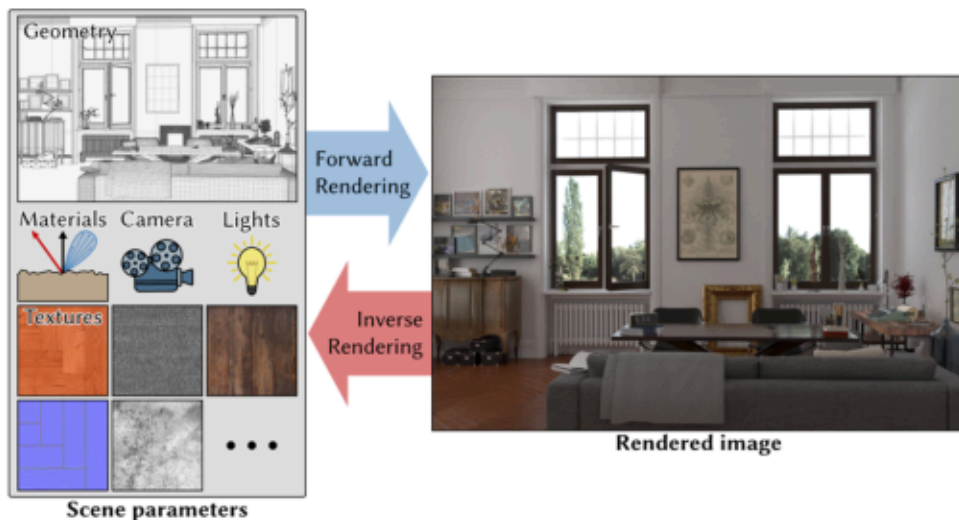


Figure 1: Inverse Rendering. Image taken from <https://tizianzeltner.com/projects/Zeltner2021Light/>

However, this problem is particularly challenging for indoor scenes. Unlike outdoor environments, indoor lighting is complex and highly variable, involving both direct light sources (like lamps) and indirect illumination caused by multiple reflections between surfaces. Accurately estimating lighting in such settings is difficult, especially when surfaces have diverse materials and when objects occlude each other, limiting visibility. These occlusions and interactions make it hard to disentangle geometry, reflectance, and lighting from images alone.

The problem becomes even more difficult in sparse-view settings, where only a few input images are available. With limited viewpoints, there is less information to constrain the reconstruction, leading to ambiguities in both geometry and lighting estimation.

In the first part of our project, we implemented the base inverse rendering approach proposed in [5]. Building on this, we extend the method by using Gaussian-based scene representations for higher fidelity of reconstruction. We show both quantitative and qualitative results and show relighting along with novel view synthesis from sparse view or single view images for multiple scenes.

2 Method

2.1 Re-Implementing the Base Paper [5]

The first part of our proposal was to implement the base paper [5] for inverse rendering. The synthetic data used in the paper for training the network is not open-sourced. Instead, we use a widely known indoor synthetic dataset Hypersim [4] for our experiments.

2.1.1 Stage - 1 : Getting the Ground Truth Lighting

The stage 1 in the paper is for estimating the ground truth environment map. Because the GT environment maps are based on distant light sources, they don't work well for indoor scenes because of occlusions. Hence the authors propose to get the GT environment map by training a neural network to predict the lighting given the GT albedo and normal to match the GT diffused image.

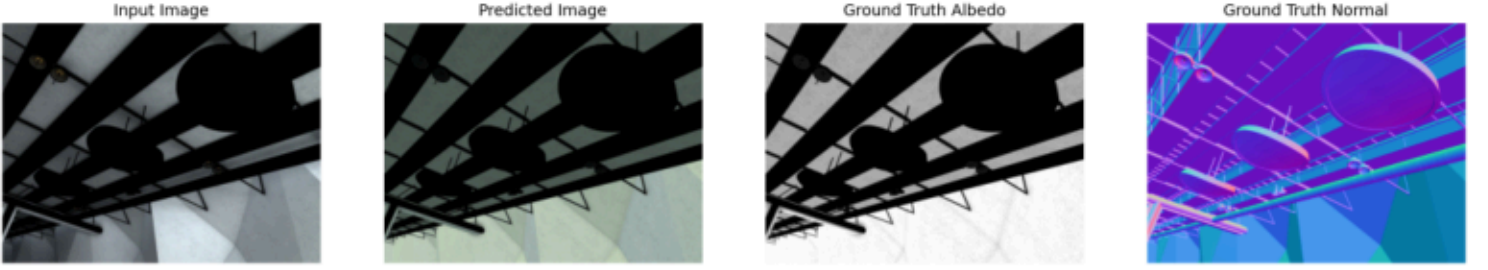


Figure 2: Predicted image after Stage 1: Getting the Ground Truth Lighting

2.1.2 Stage - 2 : Training on Synthetic Data

The next stage involves training the IRN network in the paper (for predicting albedo, normals, and environment map) on the synthetic data. This is done through direct supervision from the GT counterparts present in the synthetic data for albedo and normal, and environment map estimated from Stage - 1.



Figure 3: Predicted albedo and normal after Stage 2: Training on Synthetic data. The predicted image in this figure is from the IRN network upon passing the ground truth albedo, normal and lighting estimated from the environment map from Stage 1.

2.1.3 Stage - 3 : Training the Residual Network (RAR)

Because using the outputs in Stage - 2, we get only the diffused components, to get the full detailed image which possibly includes specular components, shadows etc, we need a residual component. This stage involves training a residual network to predict the residual image.



Figure 4: Predicted diffused, predicted residual and their combination resulting in the predicted reconstructed image after Stage 3: Training the residual Network (RAR). Here the predicted diffused and residual images are obtained by passing the predicted albedo, normal and lighting from our environment map estimation to both the IRN and RAR networks.

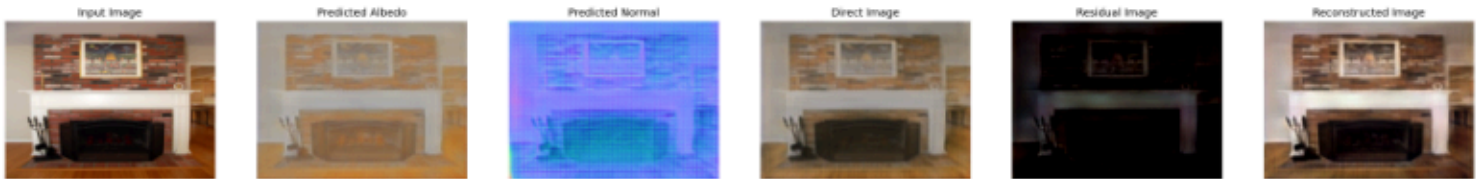


Figure 5: Predicted albedo, Predicted Normal, Direct Image, Residual Image and Reconstructed image after Stage 4: Finetuning with the IIW dataset

2.1.4 Stage - 4 : Finetuning on Real World Data

For this stage, we finetuned using the weak supervision from real world data and the RAR supervision loss, but we left out the pseudo supervision loss due to lack of details on the network used to retrieve these pseudo labels and the real world dataset didn't have these labels. We freeze the RAR network for this stage.

Finetuning with IIW Dataset While finetuning on the IIW data, which has comparison of reflectance values, we observed that our normals started getting very disrupted compared to what we had during stage 2 and 3 as shown in Figure 5.

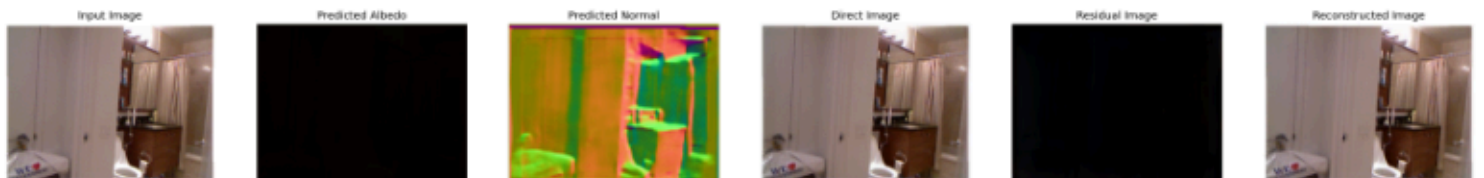


Figure 6: Predicted albedo, Predicted Normal, Direct Image, Residual Image and Reconstructed image after Stage 4: Finetuning with the NYUv2 dataset

Finetuning with NYUv2 Dataset While finetuning on the NYUv2 data, which has ground truth normals, we observed that our normals and final reconstructed image started improving, but the albedo got scaled down by a factor of 3 while lighting got scaled up by the same factor. We hypothesize that this was due to the lack of pseudo supervision. The residual images obtained from RAR were also distorted due to these inputs, due to which the reconstructed image was primarily derived from the direct image, as we can see in Figure 6.

Metric	Dataset	Ours	Paper [5]
Median Angular Error (Lower the better)	NYUv2	12.9	16.92

Table 1: Comparison of normal estimation performance on the NYUv2 dataset.

We evaluate on the NYUv2 eval dataset for the normals. The metric used in the paper for this is median angular error, where we calculate the median for the angular errors for a particular normal map with respect to GT normal map, and take its mean across the entire dataset. We get an average error of 12.9, with the one mentioned in the paper being 16.92 (Table 3). This is likely because of us using a different and more realistic synthetic dataset for training. Other tables numbers are not directly comparable because they either evaluate on their own synthetic dataset or use undisclosed samples from the real dataset which they don't mention in the paper.

2.2 Limitations of the Base Paper [5]

The reconstructed images produced by this method exhibit noticeably poor quality, as illustrated in Figure 7. This degradation can be attributed to the fact that key components, such as the albedo and normal maps, are generated by neural networks that introduce their own artifacts, which propagate into the final reconstruction. To mitigate this, we chose to move to 2D Gaussians as the scene representation instead for higher fidelity reconstruction.

2.3 Using Gaussians as Scene Representation for Inverse Rendering

We represent the scene using Gaussians to achieve high-fidelity reconstruction. However, recovering accurate geometry from sparse views is challenging due to inherent depth ambiguities. To address this, we first leverage InstantSplat [2] to obtain a reliable initial reconstruction from sparse input images. Building on this initialization, we introduce additional parameters to model lighting, and subsequently fine-tune the Gaussian representation to jointly estimate albedo and illumination.



Figure 7: **Left** : Ground Truth Image, **Right** : Reconstructed Image from [5]

2.3.1 Using InstantSplat for Base Reconstruction

We build on top of a popular sparse view novel synthesis method using Gaussian Splatting, InstantSplat [2]. The method trains a Gaussian representation using sparse views as input, and achieves high quality geometry reconstruction by leveraging priors from a pretrained foundational model for geometry reconstruction and image matching [3]. Additionally, we are using 2D Gaussians as the representation instead of 3D Gaussians, so that we can directly obtain the normal for a particular Gaussian by looking at its orientation. During the base training, we only train the DC coefficients (i.e base color) and remove the higher order SH coefficients. The base color for the j th 2D Gaussian is represented as A_j . Please refer to Fig 8 for some examples of estimated geometry from the base InstantSplat reconstruction.

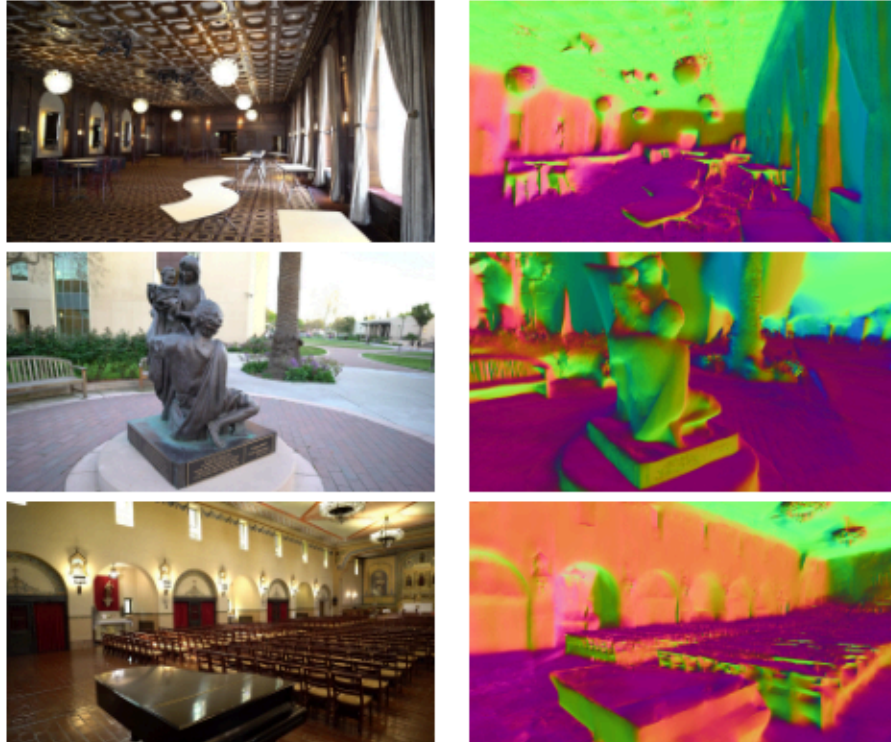


Figure 8: Ground truth (left) and rendered normals (right) for Ballroom, Family, and Church scenes using InstantSplat.

2.3.2 Recovering Albedo & Lighting

Since, we want to recover material, geometry and environment lighting in a self-supervised way from these images, we rely on accurate geometry obtained from InstantSplat obtained from base reconstruction. For now, we model the material using only its diffused component (i.e albedo) and the environment lighting using environment maps. We keep the environment map, E

which is of shape $(H_{env} \times W_{env} \times 3)$ as an optimizable tensor, and initialize it from a uniform random distribution. For each 2D Gaussian, we model its outgoing radiance as

$$c_j = \hat{A}_j \sum_{i \in H_{env} \times W_{env}} \max(0, \hat{N}_j \cdot \hat{L}_i)$$

Here, $\hat{A}_j$ is the albedo for that Gaussian obtained from the base reconstruction, $\hat{N}_j$ is its normal and $\hat{L}_i$ is the i th light source from the environment map. We set $H_{env} = 18$ and $W_{env} = 36$ as in [5]. For getting the final color of a pixel, we use the standard Gaussian Splatting rasterization, which is equivalent to ray marching and accumulating the color c_j based on the opacities. During the training, we keep E and $\hat{A}_j$ as optimizable and freeze all the other parameters of the Gaussians. $\hat{A}_j$ is initialized from the colors obtained from the base reconstruction itself. The training loss incurs a huge spike because suddenly the $\hat{A}_j$ are now replaced by c_j during the rendering and c_j has a high magnitude in the beginning. To stabilize training, we normalize c_j by dividing by a constant factor estimated using the max and min of the lighting factor $(\sum_{i \in H_{env} \times W_{env}} \max(0, \hat{N}_j \cdot \hat{L}_i))$ over all Gaussians.

2.3.3 Recovering Specular Components

We additionally add specular component according to the Blinn-Phong model as in the equation below. Here, we initialize the specular coefficients to zero, and shininess ($\alpha = 32$).

$$c_j = \hat{A}_j \sum_{i \in H_{env} \times W_{env}} \max(0, \hat{N}_j \cdot \hat{L}_i) L_i + \sum_{i \in H_{env} \times W_{env}} k_{sj} [\max(0, \hat{N}_j \cdot \hat{H}_i)]^\alpha L_i$$

3 Experiments & Quantitative Results

We evaluate the performance of single-view albedo reconstruction as well as the quality of the resulting reconstructed images on the test split of the Intrinsic Images in the Wild (IIW) [1] dataset, a standard benchmark for intrinsic image decomposition.

To assess albedo quality, we use WHDR (Weighted Human Disagreement Rate) [1], which measures consistency with human judgments; lower values indicate better alignment with ground-truth reflectance. In addition, we evaluate the fidelity of the reconstructed images using PSNR (Peak Signal-to-Noise Ratio), where higher values correspond to better reconstruction quality. As shown in Table 2, the baseline method using the input image directly as albedo yields the worst WHDR (0.5242), indicating poor intrinsic decomposition, but achieves an infinite PSNR since the reconstruction trivially matches the input image. In contrast, both learned methods significantly improve WHDR, with NIR achieving the best score (0.4087) and GS performing comparably (0.4107) for albedo estimation.

However, the reconstruction quality tells a different story. While NIR achieves a modest PSNR of 20.26, GS substantially outperforms it with a PSNR of 48.28, indicating that incorporating Gaussian-based modeling leads to much higher quality image reconstructions. This highlights a trade-off: although both methods perform similarly in albedo estimation, GS produces significantly better visual reconstructions.

Method	WHDR ↓	PSNR (Reconstructed Images) ↑
Baseline (image colors as albedo)	0.5242	∞
NIR [5]	0.4087	20.2625
GS	0.4107	48.2817

Table 2: Comparison of the base paper [5] and with using Gaussians on WHDR (for measuring albedo quality) and PSNR on subset of test set of IIW.

3.1 Qualitative Relighting Results

The relighting results for 4 different scenes are shown in Figure 9 where we use a different environment map for rendering to demonstrate inverse rendering by successfully decomposing sparse-view images into geometry (normals), material (albedo), and illumination. We see good quality images even in novel lighting conditions like that in the "Glacier" envmap, even before incorporating the specular component from Section 2.3.3.

However, we also see artifacts in more complex lighting scenarios of "Grace Cathedral" and "Ennis Brown House" renders. These artifacts appearing as discrete light patches are primarily coming from the model's current focus on direct lighting using a low-resolution environment map (18×36). Also, Because our approach currently lacks a module for indirect lighting (reflections and shadows), high-contrast light sources in the environment maps create localized high intensities that the 2D Gaussians cannot smoothly interpolate. Also, the reliance on a sparse set of views can lead to floater Gaussians or inaccurate normals in occluded areas, which can lead to these artifacts.



Figure 9: Top to Bottom: Rendered images in original lighting, "On a glacier in Banff National Forest, Canada" lighting, "Grace Cathedral, San Francisco, California" lighting, and "Dining room of the Ennis-Brown House, Los Angeles, California" lighting for Ballroom, Family and Church Scenes.

Post adding the specular component, we visualize the following results. In figure 10, we demonstrate relighting on the Ballroom scene using multiple environment maps, including a smooth interpolation from the "Dining room of the Ennis-Brown House, Los Angeles, California" (ennis) lighting to the "On a glacier in Banff National Forest, Canada" (glacier) lighting. The model produces visually coherent renderings under novel illumination conditions, indicating effective decomposition of sparse-view inputs into geometry (normals), material (albedo), and lighting. Notably, the glacier environment map, characterized by strong, high-frequency illumination, still yields plausible results, highlighting the model’s generalization capability to out-of-distribution lighting.

In figure 11, we present novel view synthesis results on the scene from the IIW dataset, where the rendered novel view closely matches the ground truth in both structure and appearance, demonstrating accurate geometry and appearance reconstruction from sparse observations.

Finally, figure 12 combines both tasks by showcasing relighting under a novel viewpoint of the Ballroom scene. The synthesized view under glacier illumination remains consistent with scene geometry while reflecting the new lighting conditions, indicating that the learned representation generalizes jointly across view and illumination changes, albeit with minor artifacts similar to those discussed earlier.

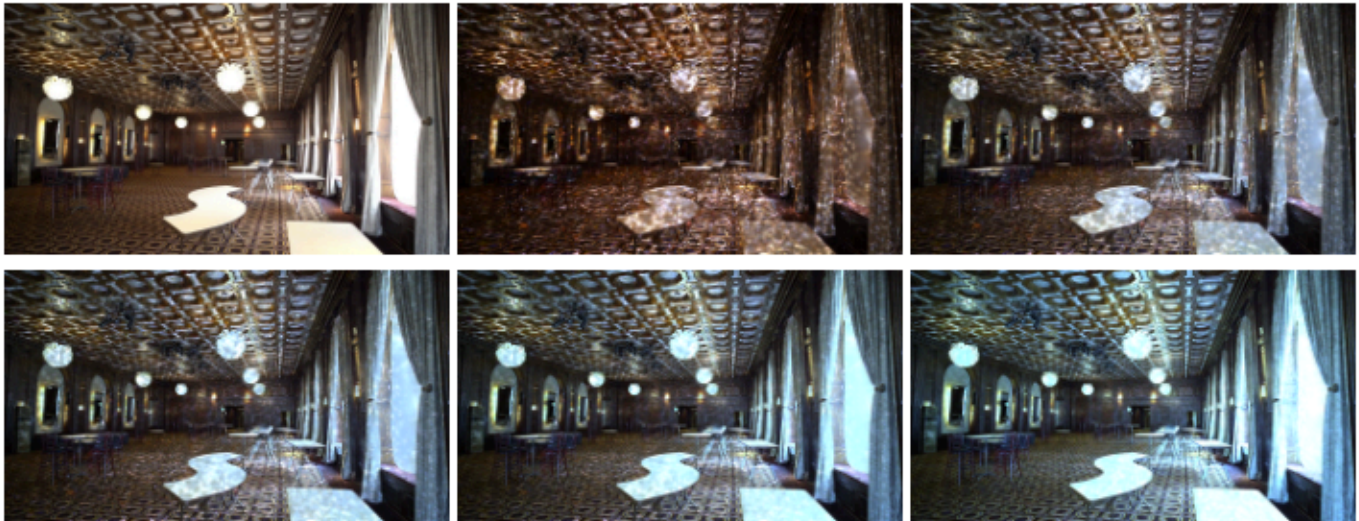


Figure 10: Ground truth image (top left), followed by interpolated renderings transitioning from "Dining room of the Ennis-Brown House, Los Angeles, California" lighting to "On a glacier in Banff National Forest, Canada" lighting for the Ballroom scene.



Figure 11: Novel view synthesis results. Left: ground truth image; right: synthesized novel view.

4 Conclusion & Future Work

In this project, we explored two complementary pipelines for inverse rendering of indoor scenes. First, we implemented a baseline approach based on prior work for single-view (and sparse-view) inverse rendering, which factorizes an input image into intrinsic



Figure 12: Novel view synthesis with relighting results. Left: ground truth image; right: synthesized novel view with glacier relighting.

components such as albedo, surface normals, and lighting. This pipeline provided a strong foundation for understanding the challenges of intrinsic decomposition, including ambiguity in lighting and the propagation of artifacts from learned per-pixel predictions.

Building on this, we proposed an extension that replaces per-pixel predictions with a structured 2D Gaussian representation inspired by recent advances in Gaussian Splatting-based scene representations. Instead of predicting albedo, normals, and lighting independently in image space, our method represents the scene as a collection of Gaussians parameterized by spatial location, covariance, surface orientation, reflectance, and illumination properties. This formulation provides a more coherent, scene-level representation that improves reconstruction quality and enables better geometric and photometric consistency across views. Empirically, while both approaches achieve comparable performance in albedo estimation, the Gaussian-based method significantly improves reconstruction fidelity, highlighting the benefits of structured representations over per-pixel predictions.

Despite these improvements, our current approach focuses primarily on modeling direct illumination. An important direction for future work is the incorporation of indirect lighting effects, such as inter-reflections, shadows, and global illumination, which are critical for realistic rendering. This could be achieved by integrating differentiable light transport formulations or extending recent Gaussian-based inverse rendering frameworks to handle multi-bounce lighting.

Future extensions could explore tighter integration with multi-view supervision, more expressive lighting models, and real-time optimization, ultimately enabling high-quality, interpretable indoor scene reconstruction that supports both relighting and novel view synthesis.

References

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